CAPSTONE PROJECT

Geospatial Imaging for Crop Suitability in Kenya

Using satellite imagery and machine learning for smarter agriculture

Navigating Agricultural Uncertainty

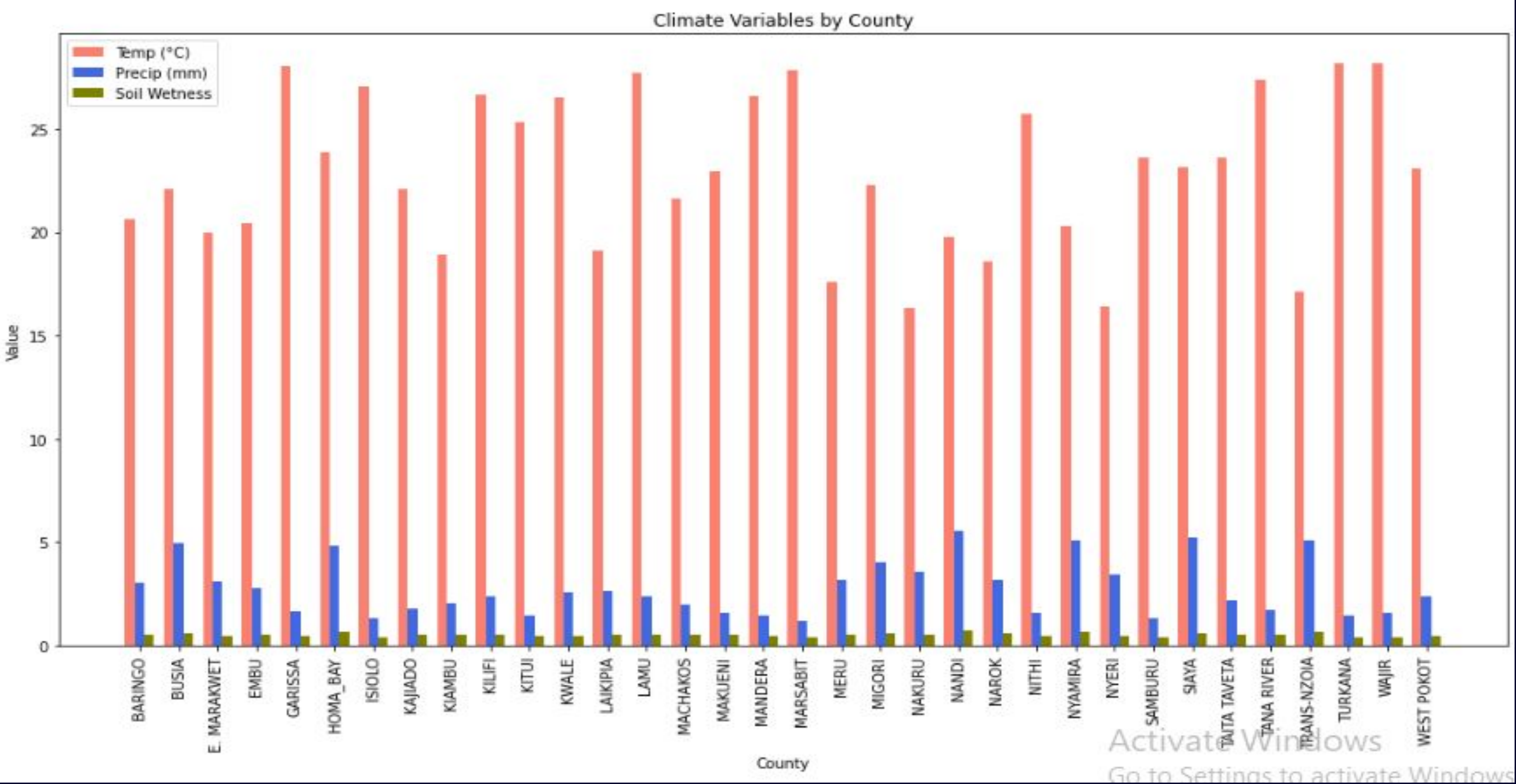
Kenyan farmers face difficulty with crop placement due to Lack of accessible data.

Navigating Agricultural Uncertainty

Lack of accessible, data results in:

- Poor decision-making
- Low crop yields
- Increased risk

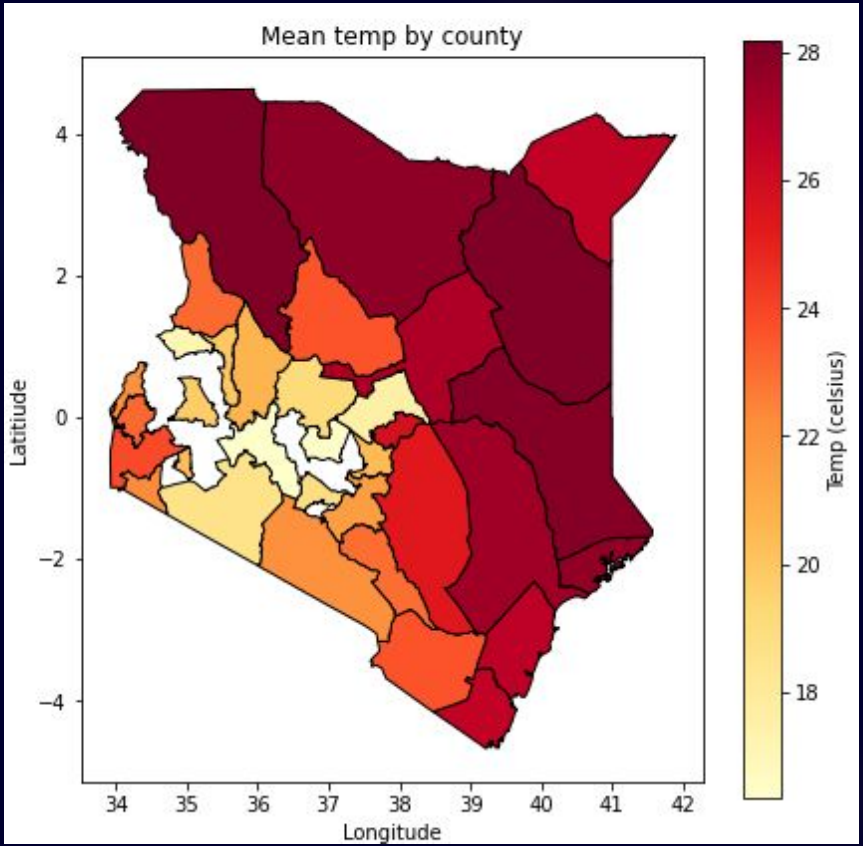
Navigating Agricultural Uncertainty



Impact of the 2024 Droughts

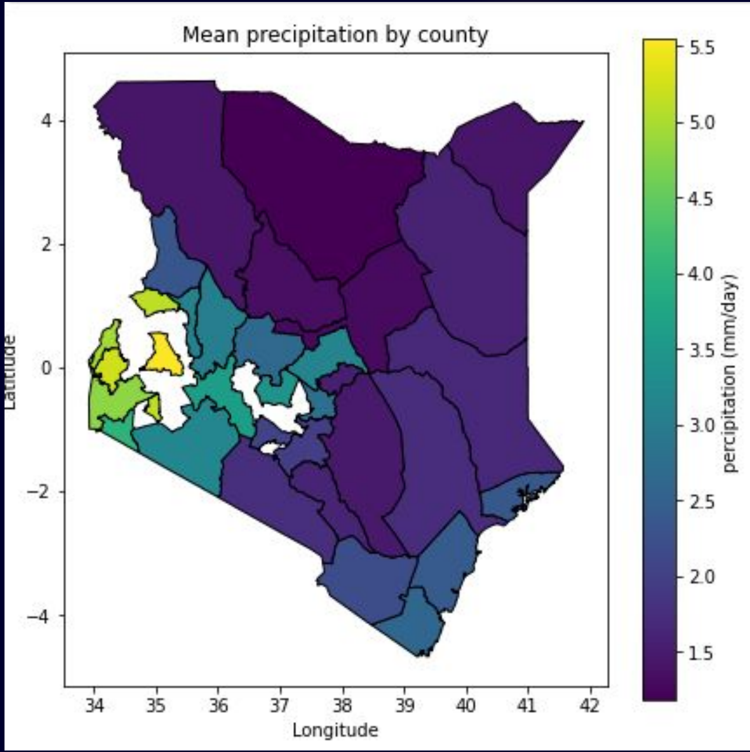
- Severe decline in agricultural production.
- Families lost:
  - Cattle
  - Crops
  - Livelihoods
- Millions of children under age five faced malnutrition

Navigating Agricultural Uncertainty



- Crop placement must be data-driven.
- Precision farming helps:
  - Maximize yields
  - Safeguard livelihoods

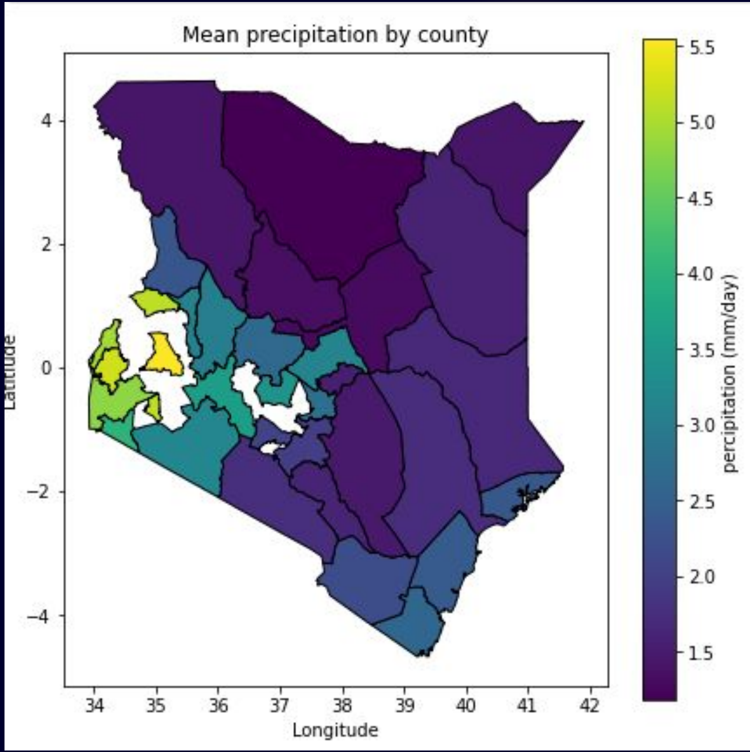
Navigating Agricultural Uncertainty



By planting at the perfect time and conditions will lead to:

- Cost savings (e.g., fertilizer, water).
- Efficiency in growth.

Navigating Agricultural Uncertainty



- Drip Irrigation: KES 75,000 – 160,000 per acre
- Sprinklers: KES 500 – 45,000 (varies by type and size)
- Rain Hoses: KES 75,000 – 110,000 per acre

Objectives

1 Enhancing Crop Suitability Mapping in Kenya

2 Develop Predictive Mapping for Kenya's Agricultural Planning

3 Model Development

Stakeholders:

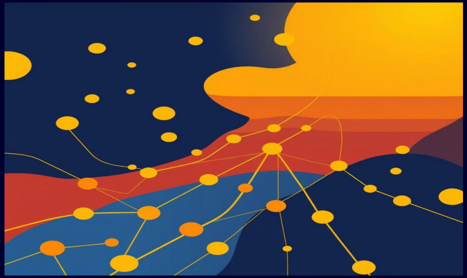
- Farmers
- Extension officers
- Ministry of Agriculture
- AgriTech Developers
- NGOs & Donors

Modeling Approach:

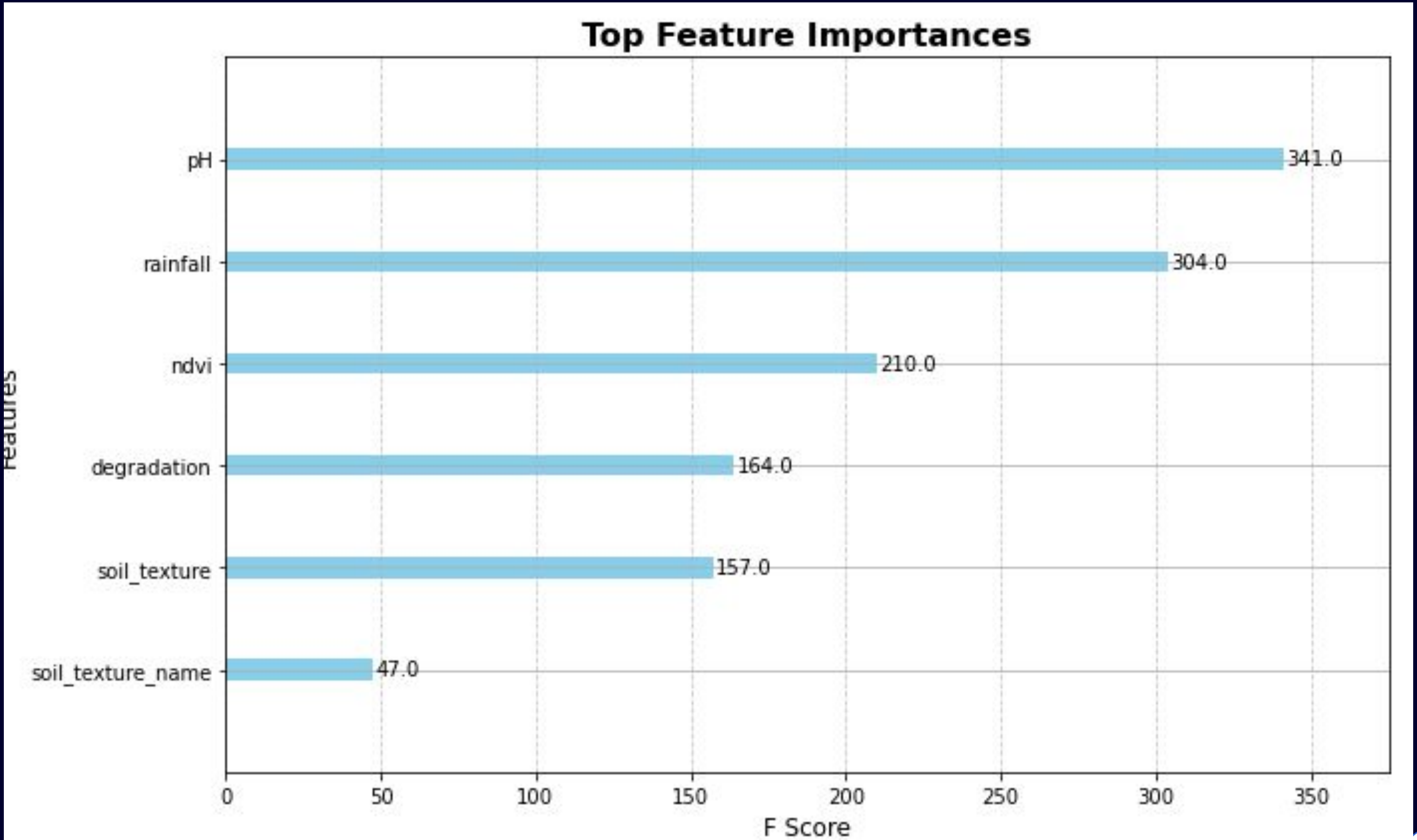
Random Forest

XGBoost

Convolutional Neural Network



Feature Importance:

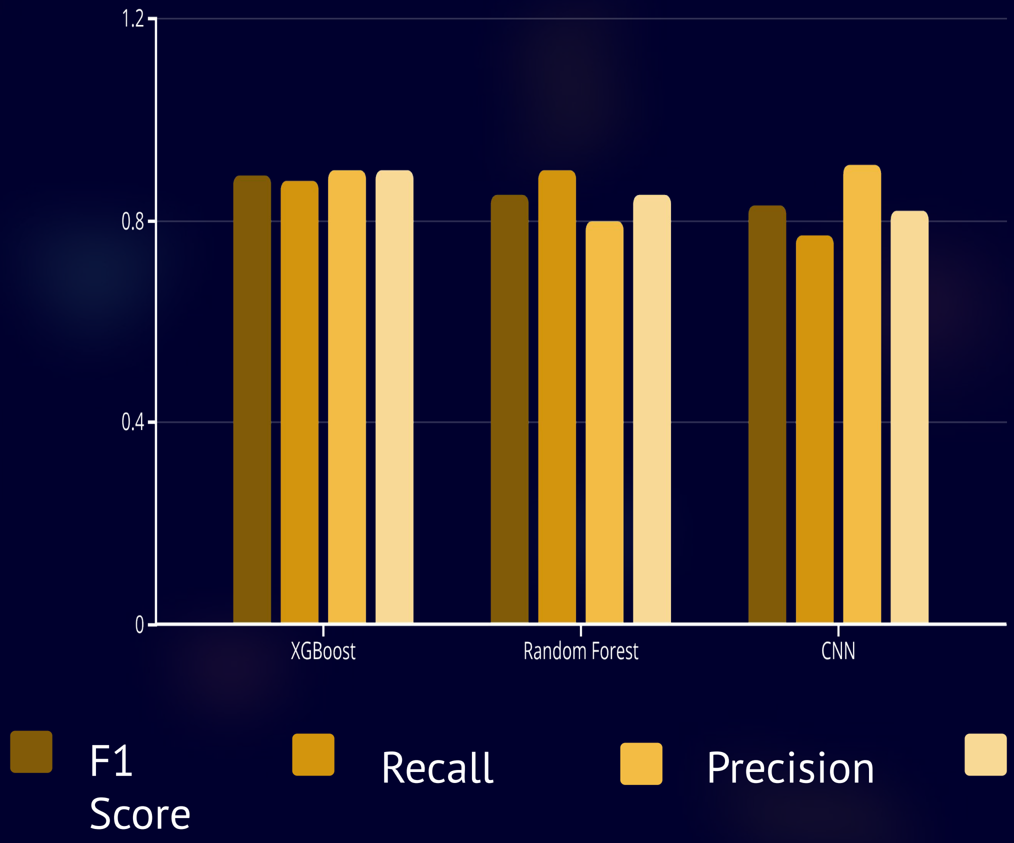


Model Comparison Results: XG Boost Leads

Model	F1 Score (Suitable)
XGBoost	**0.89**
Random Forest	0.85
CNN	0.83

- XGBoost outperformed other models with the F1 score of 0.89.
- It reliably identifies suitable areas while minimizing errors.

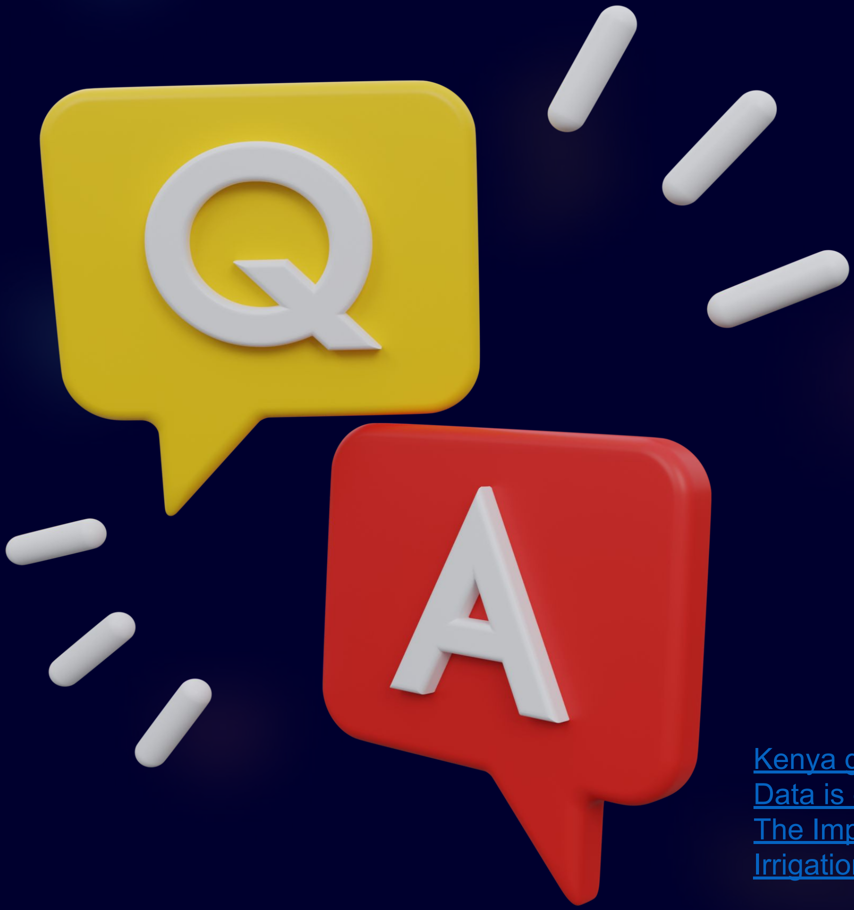
Model Comparison Results: XG Boost Leads



- XGBoost: Highest accuracy and generalization.
- Best for real-world deployment.

Recommendations

- Add multi-class classification for multiple crops
- Validate with field data and expert input
- Improve CNN with more data and augmentation



Kenya grapples with worst drought in four decades – The North Africa Post
Data is critical tool as farmers fight drought in Kenya - Kenya | ReliefWeb
The Impact of Climate Change on Crop Production in Kenya
Irrigation Farming in Kenya | 0790719020 | Aqua Hub Kenya